

What is Label Encoding?

Label Encoding converts categorical labels (like text) into numeric values (like 0, 1, 2...).

◆ Example:

["Red", "Green", "Blue"] → [2, 1, 0]

Each unique category is assigned a number.

✓ Use When:

- Categories are ordinal (have a natural order), like:
["Low", "Medium", "High"]
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✗ Avoid When:

- Categories are nominal (no order), like colors or city names — because models might treat higher numbers as more important, which is wrong.
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🔧 Tool:

- LabelEncoder from sklearn.preprocessing
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⚠ Note:

For non-ordinal categories, prefer One-Hot Encoding instead.